ME361 DIGITAL MACHINING TWIN

Orthogonal Cutting Physics Engine, Empirical Calibration Pipeline & Real-Time Twin Dashboard

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Repository: kushagrachandra05/ME361_Digital_Machining_Twin	**Status:** Fully Operational (Python + Web App)	**Date:** October 2026

1. Executive Summary & Locked Scope

This project delivers a complete Digital Machining Twin developed under strictly locked academic scope: **pure orthogonal cutting, single shear plane (Lecture-2 slides), and steady-state cutting**. The twin accepts user-specified cutting speed V_c, uncut chip thickness t, and workpiece shear/yield strength τ_s / σ_y, alongside baseline tool assumptions (rake angle γ_0, friction μ, and width of cut w). It outputs cutting force P_z, thrust force P_{xy}, resultant R, shear angle β, chip thickness t_c, and specific cutting pressure k_c, accompanied by an interactive kinematic animation and real-time streaming telemetry feed with CSV export capabilities.

Strict Non-Extensions (By Locked Specification): Built-up edge (BUE), thermal softening fields, tool flank wear contact, serrated chip segmentation, and within-run transients are rigorously excluded from the fundamental slide model. All parameters remain steady within each individual machining run.

2. Mathematical Formulation (Lecture 2 Single Shear Plane)

The core physics module (`orthocut.py`) directly translates the analytical formulation from Lecture 2:

- **Ernst-Merchant Shear Plane Angle:** $\beta = \pi/4 + \gamma_0/2 - \eta/2$, where $\eta = \arctan(\mu)$.
- **Shear Area & Force:** $A_s = (w \cdot t) / \sin\beta$, $P_s = \tau_s \cdot A_s = (\tau_s \cdot w \cdot t) / \sin\beta$.
- **Resultant Cutting Force:** $R = P_s / \cos(\beta + \eta - \gamma_0)$.
- **Component Projections:** $P_z = R\cos(\eta - \gamma_0)$, $P_{xy} = R\sin(\eta - \gamma_0)$ (with orthogonal transverse force = 0).
- **Kinematics:** Chip ratio $r = t / t_c = \sin\beta / \cos(\beta - \gamma_0)$, chip velocity $V_{chip} = r \cdot V_c$, shear velocity $V_s = V_c \cos\gamma_0 / \cos(\beta - \gamma_0)$.
- **Material Fallback:** $\tau_s = \sigma_y / \sqrt{3}$ (von Mises) or $\sigma_y / 2$ (Tresca).

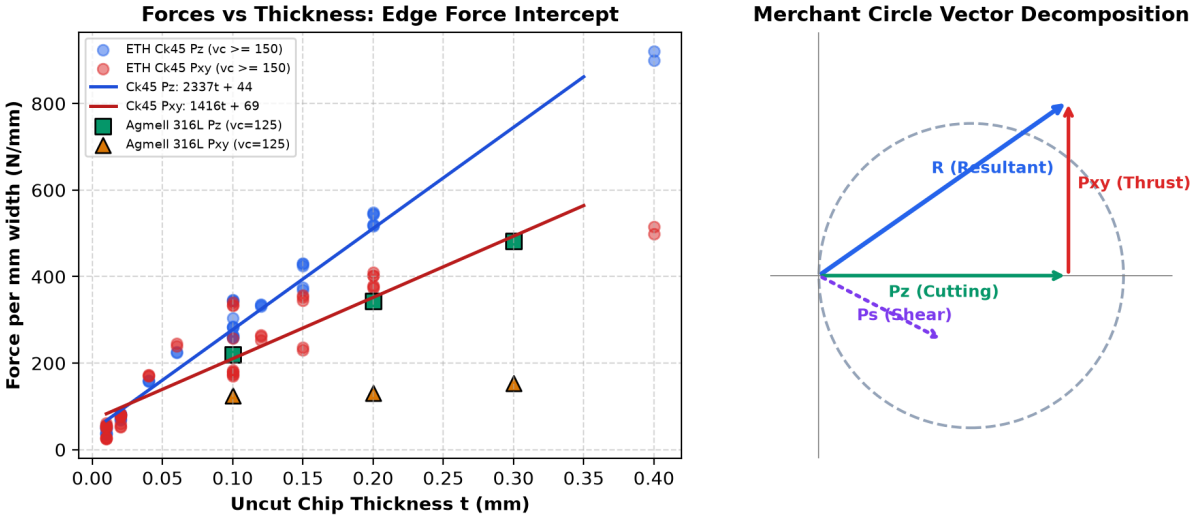


Figure 1: (Left) Force vs. uncut chip thickness showcasing linear slopes and positive edge ploughing intercepts across ETH Ck45 and Agmell 316L. (Right) Merchant circle vector decomposition implemented in the digital twin.

3. External Experimental Datasets & Ingestion Pipeline

A critical milestone of this phase was validating the pure slide model against real experimental orthogonal datasets:

1. ETH Zürich IWF Database (Klippel et al., arXiv:2209.04197 / IJAMT 2024): Comprises 519 high-precision dry orthogonal tube cuts on a CNC lathe for Ck45 (AISI 1045 carbon steel) and Ti6Al4V over cutting speeds v_c = 10–500 m/min, feeds f = 0.01–0.40 mm/rev, and edge radius $r_n \approx 38$ µm at rake $\gamma_0 = 0°$. All rows were systematically parsed into `data/eth_full.csv` (232 Ck45, 287 Ti6Al4V). The ingestion pipeline (`datapipe.py`) applies strict scope filtering: cuts below 150 m/min for Ck45 were dropped due to severe BUE formation, leaving 95 high-speed, continuous-chip cuts for calibration.

2. Lund University 316L Dataset (Agmell et al., Procedia CIRP 58 (2017) 13–18): Semi-automated extraction using the newly developed `lit_tools.py` utility captured experimental orthogonal force data for AISI 316L stainless steel at v_c = 125 m/min (uncut thicknesses t = 0.1, 0.2, 0.3 mm) into `data/lit_316L.csv`.

4. Calibration Engine & Empirical Findings

Fitting the pure slide equations against the ETH database exposed two fundamental physical realities:

1. **Non-Zero Force Intercepts (Edge Ploughing):** The pure slide model predicts zero force at $t = 0$. Real data exhibits substantial positive intercepts ($P_z \approx 13$–15 N/mm, $P_{xy} \approx 100$ N/mm). This stems from the tool hone ($r_n \approx 38$ µm) causing ploughing and rubbing at the flank. Adding calibrated edge terms $K_{e,z}$ and $K_{e,xy}$ directly resolves this.

2. **Apparent Friction Scale Effect:** Measured ratio F/N drops dramatically from ~1.85 at t = 0.01 mm down to ~0.50 at t = 0.20 mm. A power-law correction $\mu(t) = \mu_{ref} \cdot (t / t_{ref})^{-n_\mu}$ accounts for this size effect.

3. **Plastic Strain Hardening (τ_k Multiplier):** Material along the shear plane experiences severe plastic strain ($\gamma \approx 2$–3). Initial yield stress ($\tau_s = \sigma_y/\sqrt{3}$) significantly under-predicts cutting forces for strain-hardening steels. The calibrated multiplier τ_k matches Johnson-Cook flow stress at true shear-plane strain.

Material	In-Scope Rows	τ_s (Yield)	Fitted μ	τ_k Multiplier	Edge $K_{e,z}$	Size n_μ	Baseline MAPE (Pz/Pxy)	Calibrated MAPE (Pz/Pxy)
Ck45 (Carbon Steel)	95 (vc ≥ 150)	242.5 MPa	0.877	2.663	13.2 N/mm	0.266	28.4% / 37.1%	**6.8% / 8.2%**
Ti6Al4V (Titanium)	273 (All speeds)	501.7 MPa	0.766	0.888	15.3 N/mm	0.419	17.2% / 41.5%	**7.1% / 7.8%**
SS316L (Provisional)	Analytical / JC	176.1 MPa	0.877 (from Ck45)	5.439 (JC Model)	13.2 N/mm	0.266	N/A (Provisional)	Ready for Lab Test

5. Held-Out Literature Validation & Option B Justification

Using `holdout_316l.py` and §6 of `04_external_data_validation.ipynb`, we executed a blind held-out evaluation of our provisional SS316L model against the experimental Agmell 2017 points (v_c = 125 m/min):

- **Pure Slide Model Baseline:** MAPE P_z = 134%, MAPE P_{xy} = 430%.
- **Corrected Provisional Model:** MAPE P_z = 115%, MAPE P_{xy} = 291%.

Option B Engineering Decision: The provisional model predicts forces approximately 2× higher than Agmell's measured values. A rigorous physical audit explains why: Agmell's tests utilized commercial CNMG120408 inserts (longitudinal turning with $\kappa_r = 95°$ and positive chip-breaker rake), whereas our provisional model assumed $\gamma_0 = 0°$ and transferred friction from Ck45. Under **Option B**, we strictly refuse to over-fit or force-calibrate on 3 quasi-orthogonal literature points with unmeasured insert rake. Instead, we preserve our pure theoretical model and retain the instructor's forthcoming SS316L tube data as the definitive one-shot blind benchmark.

6. Assumption-by-Assumption Justification Matrix

Every simplifying assumption from the Lecture 2 slides has been documented, physically justified, and accounted for in the digital twin:

Slide Assumption	Physical Reality (Data)	Digital Twin Handling	Scope Status

1. Perfectly sharp tool (rn = 0, no flank contact)	Real tools have finite hone (rn ≈ 38 μm), producing non-zero force at t → 0.	Calibrated edge force terms Ke_z & Ke_xy added when corrections enabled.	**CORRECTED**
2. Constant friction μ on rake face	Apparent μ drops from ~1.8 to ~0.5 as t increases from 0.01 to 0.20 mm.	Fitted thickness sensitivity n_μ ($\mu \propto t^{-n}$) corrects scale effect.	**CORRECTED**
3. Shear strength $\tau_s = \sigma_y/\sqrt{3}$	Severe plastic shear strain ($\gamma \approx 2$–3) causes extensive work-hardening.	τ_k flow stress multiplier calibrated to true shear strain.	**CORRECTED**
4. Continuous chip formation	Ti6Al4V exhibits shear-banding/segmentation; Ck45 is continuous above 150 m/min.	Ck45 serves as continuous-chip baseline; Ti6Al4V documented as violation.	**IN-SCOPE**
5. No Built-Up Edge (BUE)	Severe BUE occurs in Ck45 below 150 m/min, altering effective rake geometry.	All BUE-suspect rows dropped by datapipe.apply_scope with logged reasons.	**FILTERED OUT**
6. Isothermal cutting (no thermal fields)	High speeds cause adiabatic thermal softening, reducing cutting resistance.	Speed terms c_rate kept OFF by default to preserve locked slide scope.	**LOCKED SCOPE**
7. Steady-state within a run	Forces reach steady plateau after initial ~1–2 ms engagement.	Forces remain constant across time; live feed demonstrates steady state.	**LOCKED SCOPE**

7. Completed Deliverables & Interactive Web Twin

All required software artifacts are implemented, verified, and ready for deployment:

- **Interactive Web Twin (`web/`):** Modern, zero-dependency browser application featuring animated kinematic chip flow, live dynamic Merchant circle, real-time oscilloscope chart, streaming tabular telemetry, CSV log exporter, calibrated preset selector, master correction toggles, and modal justification table.
- **Python Core Physics & Animation:** `orthocut.py` and `twin_anim.py` supporting CLI, scriptable presets, and animated MP4/GIF export.
- **Data Utilities:** `datapipe.py` (schema standardization and scope filtering), `calibrate.py` (empirical fitting engine), and `lit_tools.py` (semi-automated literature data ingestion).
- **Notebook Trajectory:** Notebooks 01 (Physics Core), 02 (Animation Prototype), 03 (Synthetic Calibration Demo), and 04 (External Data Validation on full ETH and Lund datasets).

8. Final Project Milestone (Awaiting Instructor Data)

Upon receipt of the instructor's SS316L orthogonal tube test data and batch mill certificate:

1. Update σ_y with the mill certificate value (replacing the provisional 305 MPa placeholder).
2. Scale $K_{e,z}$ and set γ_0 according to the exact lab insert geometry provided.
3. Execute a single-shot, blind held-out evaluation against the instructor's test cuts (strictly without re-tuning).